

B. Tech Degree IV Semester (Special Supplementary) Examination, March 2007

ME 403 APPLIED THERMODYNAMICS

(2002 Admissions onwards)

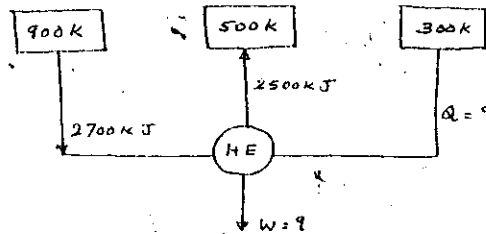
Time : 3 Hours

Maximum Marks : 100

- I. (a) Explain how the first law of thermodynamics is applied to a control volume. (10)
 (b) 0.5Kg of an ideal gas expands adiabatically until its pressure is halved. During expansion, the gas does 30KJ of external work and its temperature falls from 500K to 410K. Make calculations for the adiabatic exponent and the characteristic gas constant. (10)

OR

- II. (a) State the Kelvin-Planck and Clausius statements of the second law of thermodynamics and establish the equivalence between them. (10)
 (b) A reversible heat engine shown in the figure absorbs 2700KJ from a reservoir at 900K and rejects 2500KJ to a reservoir at 500K. Determine the heat interaction with the reservoir at 300K and the net work delivered by the engine. (10)



- III. (a) State Dalton's law and Amagat's law of mixture of ideal gases. (5)
 (b) Derive an expression for the equivalent Molecular weight of a mixture of ideal gases. (5)
 (c) Determine the mole fractions of a gas mixture that consists of 60% CH₄ and 40% CO₂ by mass. Also determine the gas constant of the mixture. (10)

OR

- IV. (a) Derive an expression for the characteristic gas constant of a mixture of ideal gases. (10)
 (b) Consider a gas mixture that consists of 3Kg of O₂, 5Kg of N₂ and 12Kg of CH₄. Determine
 (i) the mass fraction of each component.
 (ii) the mole fraction of each component.
 (iii) the average molar mass and gas constant of mixture. (10)

- V. A boiler is fired with a fuel having a mass composition C = 86.1%, H₂ = 3.9%, O₂ = 1.4%, ash = 8.6%. Volumetric analysis of dry flue gases gives CO₂=12.7%, CO=1.4%, O₂=4.1%, N₂=81.8%. Calculate:
 (i) The total amount of flue gas per kg of fuel fired. (10)
 (ii) The percentage of excess air supplied above the minimum quantity required for complete combustion of fuel. (10)

OR

- VI. (a) What is enthalpy of formation? How does it differ from the enthalpy of combustion? (10)
 (b) Explain what is meant by adiabatic flame temperature. A fuel is completely burned first with the stoichiometric amount of air and then with the stoichiometric amount of pure oxygen. For which case will the adiabatic flame temperature be higher? Why? (10)
- VII. (a) With a neat sketch explain the valve timing diagram of a 4 stroke CI engine. (10)
 (b) A petrol engine has a cylinder diameter of 60mm and stroke 100mm. If the mass of the charge admitted per cycle is 0.0002Kg, find the volumetric efficiency of the engine. Assume characteristic constant for the charge is 287.Nm/KgK. (10)

OR

- VIII. (a) An IC engine uses 6Kg of fuel having calorific value 42000KJ/Kg in one hour. The indicated power developed is 20KW. The temperature of 11.5Kg of cooling water was found to rise through 25°C per minute. The temperature of 4.2Kg of exhaust gas with specific heat 1.005KJ/Kg was found to rise through 220°C. Prepare a heat balance sheet of the engine. (10)
 (b) Discuss the various factors affecting the performance of a CI engine. (10)

OR

- IX. (a) What are the basic components of a lubricating system? Explain its functions briefly. (10)
 (b) With neat sketch explain the working of Magneto-ignition system of SI engine. (10)
- X. (a) What are the general objectives in the design of combustion chamber of SI engines. (10)
 (b) Discuss the phenomenon of knock in CI engine. How do the injection timing and fuel quality affect the engine knock? (10)